

USER GUIDE

APIC ProtoHub

How to use the platform

This guide is in two parts. Part A is for innovators — students, startups, and anyone who wants to make a prototype. Part B is for centre staff who run the machines. Find your part and follow the steps.

IF YOU ARE...	YOU WANT TO...	GO TO
An innovator / student	Find a machine and book it	Part A (page 2)
Centre staff / manager	Price jobs and run the queue	Part B (page 4)

BEFORE YOU START. Open the platform in any web browser on a phone or computer. Innovators open app.html; centre staff open manager.html. No installation, no account setup for this version — just open the file and go.

For innovators — find and book a machine

You have a design and you need it made. You do not need to know which machine or which centre — the platform figures that out from your part. Here is the whole journey, start to finish.

The idea in one line

Describe your part → see machines that can make it → book a time → track it until it is ready. You never pick a machine blind; the platform only shows you equipment that can physically do your job.

Step by step

1 Enter your part size

At the top of the screen, type your part's dimensions into the three boxes — **length × width × height**, in millimetres. Not sure of exact size? Use the largest measurement in each direction. The platform rotates the part automatically to see if it fits.

2 Choose your material

Pick from the **Material** dropdown — PLA, ABS, Nylon, Aluminium, Acrylic, and more. If you don't mind, leave it on "Any material" to see every option.

3 Set how precise it needs to be

The **Precision needed** dropdown ranges from "Not fussy" to "Very precise (±0.02mm)". A rough bracket needs low precision; a part that must fit exactly needs high. Higher precision means fewer machines qualify.

4 Tell it who you are and where

Set **You are** to Student, Startup, Industry, or Individual — students get discounted rates automatically. Set **Near** to your nearest city so the closest machines show first.

5 Press Find machines

Click **Find machines**. You'll see a list of machines that can make your part, best match at the top. Each card shows the machine, the centre, how far away it is, the hourly rate (with your discount already applied), and whether your material is in stock.

NOTHING SHOWS UP? That means no machine in the network can make the part as described. Try a smaller size, a different material, or a looser precision setting. The platform never shows you a machine that can't do the job — so an empty result is honest, not broken.

6 Book a time

On the machine you want, press **Book**. A calendar opens showing free time slots for the next few days — booked and past times are greyed out. Pick a slot. You'll see the estimated cost (machine time + setup, with your discount).

7 Confirm

Press **Confirm booking**. Your slot is held and a job number is created (e.g. JOB-1234). Its status is "Quote pending" — the centre reviews your file and confirms the exact price before any work or charge happens.

What happens after you book

Your job moves through clear stages. You can watch each one:

STAGE	WHAT IT MEANS
Quote pending	Centre is reviewing your file to set the price
Quote sent	Price is ready — accept it to proceed
Queued	Accepted and waiting its turn on the machine
Running	Your part is being made right now
Ready	Done — go collect it from the centre

For centre staff — run the queue

The manager console is your shop-floor board. It shows every job that needs attention, most urgent first, and tells you the one thing to do next on each. You never hunt through a register or field status calls — the screen is the single source of truth.

Your home screen

When you open the console you see three things:

- **The stat strip (top):** today's numbers — jobs awaiting a quote, active jobs, jobs ready for pickup, machines up, machines down, and low-stock warnings.
- **The job queue (centre):** a stack of job cards ordered by urgency. Jobs needing a price float to the top; each card names the innovator, machine, material, and quantity.
- **The side panels (right):** material stock levels with low-stock warnings, and each machine's status (available, in use, maintenance).

The daily workflow

1 Price the new jobs

Jobs marked **Needs quote** sit at the top. Press **Price this job**. Enter the machine time (minutes) and material (grams). The price calculates live — machine cost + material + setup, minus any student discount — showing the full breakdown. Press **Send quote** to send it to the innovator.

2 Move accepted jobs through the machine

Once an innovator accepts, the job becomes ready to run. Each card shows one button for the next step, in order: **Start setup** → **Begin run** → **Move to QC** → **Pass QC** → **Mark collected**. Press each as you complete that stage. The innovator sees the same progress live.

MATERIAL IS DEDUCTED AUTOMATICALLY. When you press **Begin run**, the material for that job is subtracted from stock. If there isn't enough, the system blocks the run and tells you — so a job never starts on filament you don't have.

3 Handle problems

If a print fails or a part is scrapped, press **Fail** on that job to take it out of the active queue and record what happened.

4 Keep material topped up

The stock panel warns you when a material drops below its reorder level (the bar turns red). After refilling a spool, press **+ Stock** to record it. This keeps the "low stock" count honest so jobs don't fail mid-run.

Why every action is safe

The console is built so you cannot make the common mistakes:

- You cannot double-book a machine — the system rejects overlapping slots outright.
- You cannot skip a step — a job can't jump from queued to done; it goes in order.
- You cannot start a run without enough material — the system stops you and says why.
- Every action is logged with your name and the time, so the centre's history is always complete.

If something looks wrong

WHAT YOU SEE	WHAT TO DO
"No machine can make this part"	Expected when the part is too big, the material is unavailable, or precision is too tight. Loosen one of them.
A time slot is greyed out	It is already booked or in the past. Pick another. This is normal.
"Insufficient stock" when starting a run	The machine doesn't have enough material. Refill and press + Stock, then start the run.
A button you expect isn't there	Actions appear only when valid for that job's stage — e.g. "Pass QC" shows only during QC.

About this version

Maturity: this is a validated prototype (TRL 4). The platform's core — matching, booking, pricing, and job tracking — is built and stress-tested: double-booking and step-skipping are provably impossible, and pricing is verified to the paise. What it does not yet have is real centre data and live accounts; those come with a pilot at a real APIC.

For this version, **the machines, rates, and centres shown are realistic samples for demonstration.** A live deployment replaces them with one real centre's actual equipment and price card.